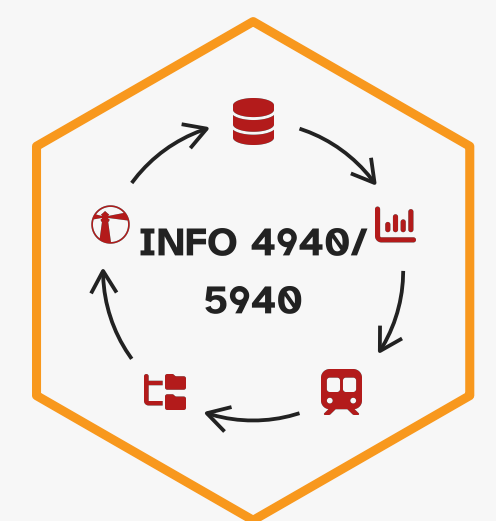




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HW 02 - Build better data

HOMEWORK

MODIFIED

September 11, 2025

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Important

This homework is due September 17 at 11:59pm ET.

Learning objectives

- Define and fit machine learning models
- Define data preprocessing and feature engineering steps for predictive models
- Estimate a range of model types
- Utilize exploratory analysis and domain expertise to inform feature engineering and model selection

Getting started

- Go to the [info4940-fa25](#) organization on GitHub. Click on the repo with the prefix **hw-02**. It contains the starter documents you need to complete the lab.
- Clone the repo and start a new workspace in Positron. See the [Homework 0 instructions](#) for details on cloning a repo and starting a new R project.

General guidance

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Guidelines + tips

- Set your random seed to ensure reproducible results.
- Use [caching](#) to speed up the rendering process.
- Use **parallel processing** to speed up rendering time. Note that this works differently on different systems and operating systems, and it also makes it harder to debug code and track progress in model fitting. Use at your own discretion.

Remember that continuing to develop a sound workflow for reproducible data analysis is important as you complete this homework and other assignments in this course. There will be periodic reminders in this assignment to remind you to render, commit, and push your changes to GitHub. You should have **at least 3 commits with meaningful commit messages** by the end of the assignment.

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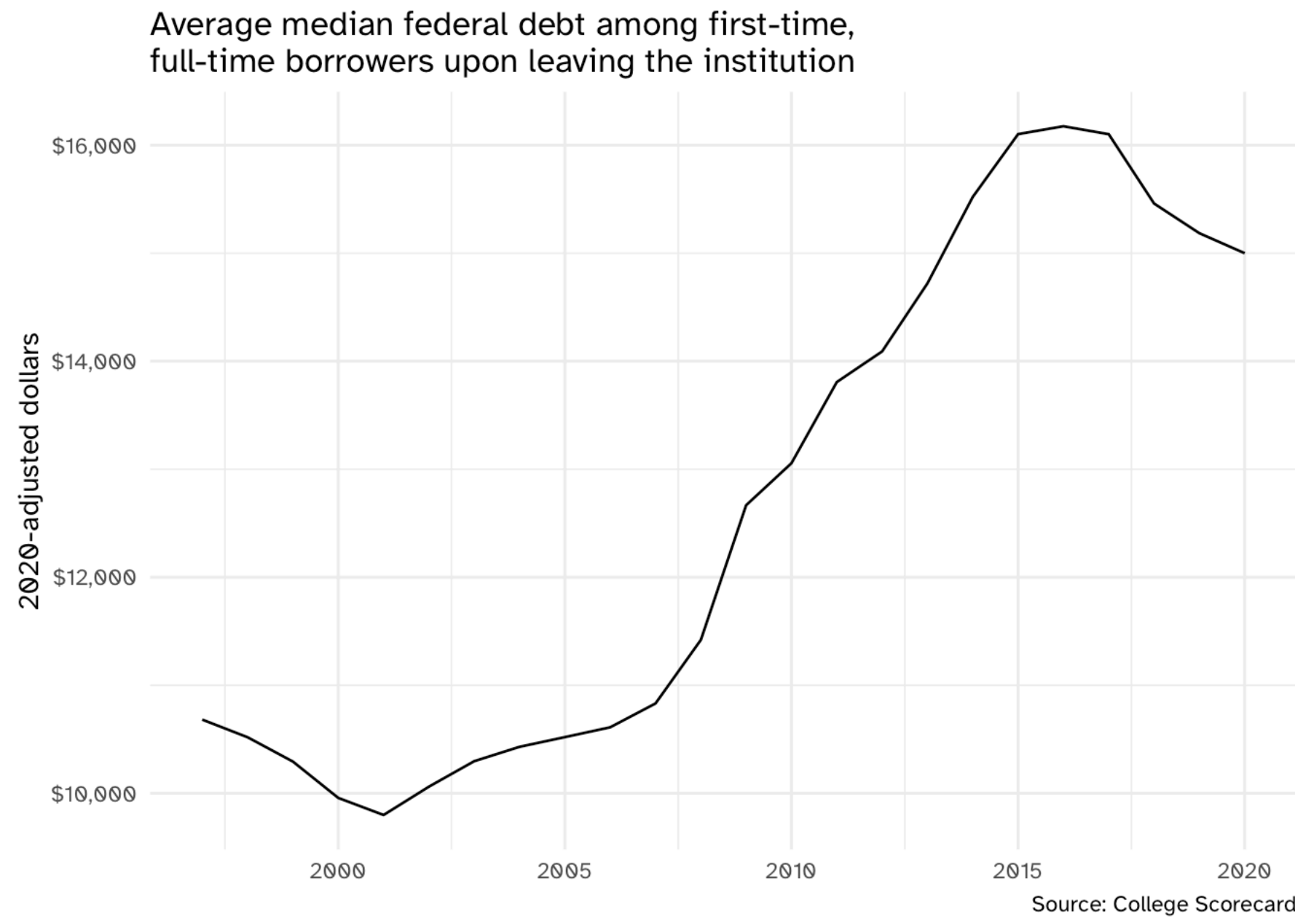
Workflow + formatting

Make sure to

- Update author name on your document.
- Label all code chunks informatively and concisely.
- Follow consistent code style guidelines.
- Make at least 3 commits.
- Resize figures where needed, avoid tiny or huge plots.
- Turn in an organized, well formatted document.

Student debt in the United States

Median student debt in the United States has increased substantially over the past twenty years.



data/scorecard.csv contains a portion of the database¹ for academic year 2022-23 with the following variables:

Column name	Variable description
unit_id	Unit ID for institution
name	Institution name
state	State postcode
act_med	Median ACT score for admitted students
adm_rate	Admission rate
comp_rate	Completion rate for first-time, full-time students
cost_net	The average annual total cost of attendance, including tuition and fees, books and supplies, and living expenses for all full-time, first-time, degree/certificate-seeking undergraduates, minus the average grant/scholarship aid
cost_sticker	The average annual total cost of attendance, including tuition and fees, books and supplies, and living expenses for all full-time, first-time, degree/certificate-seeking undergraduates
death_pct	Percent of student who died within 4 years at original institution
debt	Median debt of student graduating in 2022-23
earnings_med	Median earnings of students working and not enrolled 6 years after entry
earnings_sd	Standard deviation of earnings of students working and not enrolled 6 years after entry
faculty_salary_mean	Average faculty salary
faculty_full_time	Proportion of faculty that are full-time
female	Share of female students
first_gen	Share of first-generation students
instruct_exp	Instructional expenditures per full-time student
locale	Locale of institution
median_hh_inc	Median household income of students
open_admissions	Open admissions policy
pbi	Predominantly black institution
pell_pct	Percentage of student who receive a Pell grant
retention_rate	First-time, full-time student retention rate
sat_mean	Average (mean) SAT score of admitted students
test_policy	Test score requirements for admission
type	Type of institution
veteran	Share of veteran students
women_only	Women-only college

Our goal is to **predict the median debt of students graduating in 2022-23**.

Exercise 1

Import and partition the data. Import the data and reproducibly partition it into training and test sets. Briefly justify your choice of partitioning strategy.

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Remove rows with missing values for debt

Prior to partitioning the data, remove any rows with missing values for the outcome variable `debt`.² You may leave rows with missing values for predictor variables in the dataset.

This is an example of a data preprocessing action that actually **should** be done before partitioning the data, since it involves the outcome variable.

Exercise 2

Inspect the distribution of the outcome variable `debt` and conduct exploratory analysis with potential features of interest. Use visualizations and summary statistics to explore the data focusing on the outcome of interest and at least five other variables. Summarize your findings.

Exercise 3

Resample the training data and fit a null model. Resample the training data using an appropriate method (justify your choice) and fit a null model to establish a baseline of comparison. Interpret the results of the null model.

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Metrics for evaluation

For all models in this assignment, estimate and interpret the **root mean squared error (RMSE)** and **mean absolute error (MAE)**.

Exercise 4

Train a basic linear regression model. Train a linear regression model using all features in the dataset. Be sure to exclude `unit_id` and `name` as predictors since they are ID columns and have no predictive value. Implement the following data preprocessing steps as part of your modeling workflow:

- Impute missing values with arithmetic means/modes for numerical/nominal predictors, respectively
- Convert categorical variables to dummy variables
- Remove zero-variance predictors
- Remove highly collinear predictors

Evaluate the model's performance and compare it to the null model.

Exercise 5

Use feature engineering to improve the linear regression model. Use your exploratory analysis from Exercise 2 and domain expertise to inform additional data preprocessing/feature engineering steps that may improve model performance. Document why you decided to implement these steps. Implement these steps as part of your modeling workflow, and evaluate the model's performance. Report your improvement over the basic linear regression model from Exercise 4 as a percentage reduction in RMSE and MAE.

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How much do I need to do?

Ultimately the goal is to improve the model's performance and reduce the RMSE and MAE. There is no minimum number of feature engineering steps you need to implement, but be warned: minimal effort will likely result in minimal improvement and a low grade for the exercise. You will be evaluated in part on both your effort and the improvement you achieve.

Exercise 6

Evaluate a random forest model. Train a random forest model. Implement any required data preprocessing but do not perform any feature engineering. Evaluate the model's performance and compare it to the linear regression models from Exercises 4 and 5.

Exercise 7

Finalize your model and evaluate on the test set. Pick the best model (explain why you decided it is the best). Train it using the entire training set (not resampled) and evaluate it using the test set. How does it compare to the null model? What about the metric from the resampled training set? How effective do you find this model to be?

Generative AI (GAI) self-reflection

As stated in the [syllabus](#), include a written reflection for this assignment of how you used GAI tools (e.g. what tools you used, how you used them to assist you with writing code), what skills you believe you acquired, and how you believe you demonstrated mastery of the learning objectives.

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Render, commit, and push one last time.

✅

Make sure that you commit and push all changed documents and your Git pane is completely empty before proceeding.

Wrap up

Submission

- Go to <http://www.gradescope.com> and click *Log in* in the top right corner.
- Click *School Credentials* → *Cornell University NetID* and log in using your NetID credentials.
- Click on your *INFO 4940/5940* course.
- Click on the assignment, and you'll be prompted to submit it.
- Mark all the pages associated with exercise. All the pages of your homework should be associated with at least one question (i.e., should be “checked”).

Grading

- Exercise 1: 2 points
- Exercise 1: 10 points
- Exercise 3: 4 points
- Exercise 4: 8 points
- Exercise 5: 16 points
- Exercise 6: 4 points
- Exercise 7: 6 points
- GAI self-reflection: 0 points
- Total: 50 points

Footnotes

← HW 01 - Make a model

HW 03 - Tune and evaluate models →